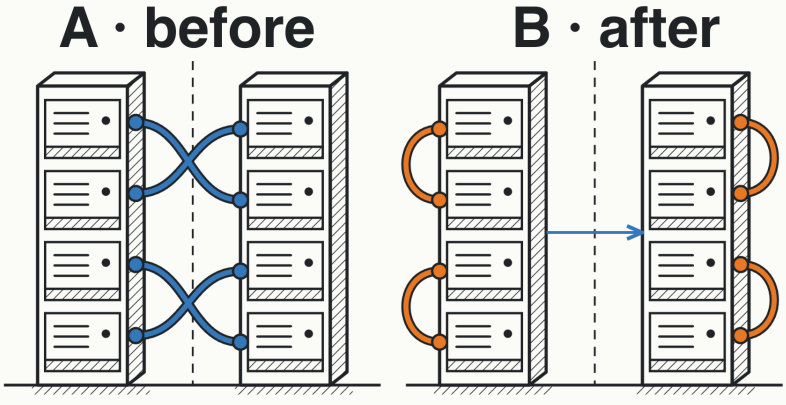
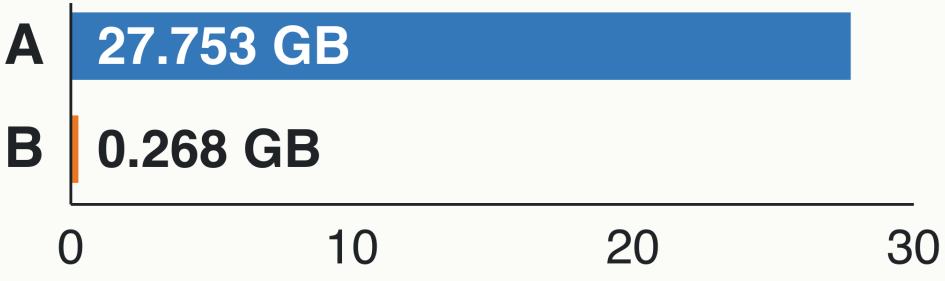
Placement moves rack traffic

Same 64 GPUs · hypothetical update



Cross-rack GB · both directions



Within racks: 0.805 → 28.290 GB
Total off-node payload unchanged

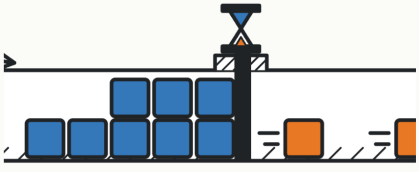
Two ordered lower bounds

Hypothetical plan · ms per update

A before · B after

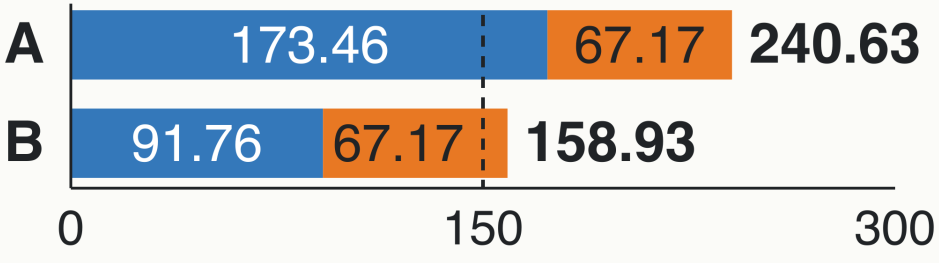
Pre-clip

Post-clip



Clipping barrier

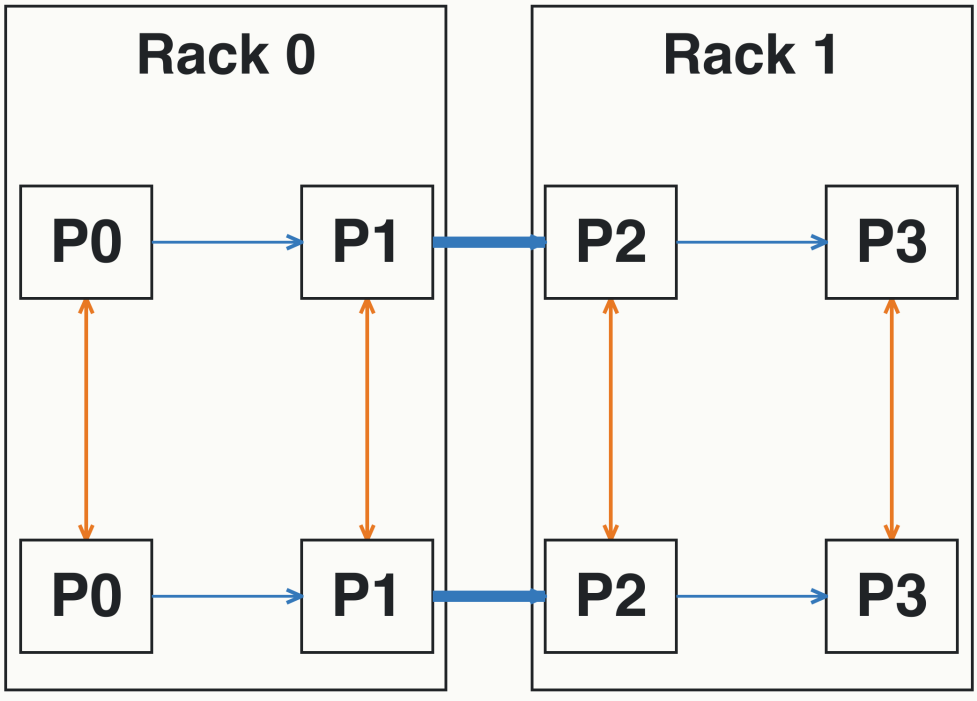
Illustrative 150 ms budget



Full DP all-reduce before norm
FP32 Muon · one attention owner
Assumed 100 TFLOP/s per GPU
Rack 80 / node 40 GB/s per direction
Balanced routing · not runtimes

Pair stages inside racks

B after · Replica 0 top · Replica 1 bottom



P1 → P2 crosses racks

DP inside racks · 8 GPUs per node